

Problem 1

- (a) For all $n \in \mathbb{N}$, let $P(n)$ be the statement that $x_{n+1} > x_n$.

Base case. When $n = 1$, we have

$$x_1 = 0, \quad x_2 = \sqrt{x_1 + 10} = \sqrt{0 + 10} = \sqrt{10}.$$

Since $\sqrt{10} > 0 = x_1$, the statement $P(1)$ is true.

Inductive hypothesis. Assume that $P(n)$ is true for some $n \in \mathbb{N}$, that is,

$$x_{n+1} > x_n.$$

Inductive step. Adding 10 to both sides gives

$$x_{n+1} + 10 > x_n + 10.$$

Taking square roots of both sides,

$$\sqrt{x_{n+1} + 10} > \sqrt{x_n + 10}.$$

By the definition of the sequence,

$$x_{n+2} > x_{n+1},$$

so $P(n+1)$ is true.

Conclusion. Since $P(1)$ is true and $P(n) \Rightarrow P(n+1)$, it follows that

$$P(n) \text{ is true for all } n \in \mathbb{N}.$$

- (b) Let $P(n)$ be the statement that $x_{n+1} < x_n$ for all $n \in \mathbb{N}$.

Base case. When $n = 1$, we have

$$x_1 = 10 = 2 \cdot 5, \quad x_2 = \sqrt{x_1 + 10} = \sqrt{20} = 2\sqrt{5}.$$

Since $2\sqrt{5} < 10$, we have $x_2 < x_1$, so $P(1)$ is true.

Inductive hypothesis. Assume that $P(n)$ is true for some $n \in \mathbb{N}$, that is,

$$x_{n+1} < x_n.$$

Inductive step. Adding 10 to both sides gives

$$x_{n+1} + 10 < x_n + 10.$$

Taking square roots of both sides,

$$\sqrt{x_{n+1} + 10} < \sqrt{x_n + 10}.$$

By the definition of the sequence,

$$x_{n+2} < x_{n+1},$$

so $P(n+1)$ is true.

Conclusion. Since $P(1)$ is true and $P(n) \Rightarrow P(n+1)$, it follows that

$$P(n) \text{ is true for all } n \in \mathbb{N}.$$

Problem 2

Consider the polynomial

$$f(x) = 10x^4 + 7x^3 - 4x^2 + 14x + 3.$$

By the Rational Root Theorem, any rational root has the form

$$x = \frac{a}{b},$$

where $\gcd(a, b) = 1$, $a \mid 3$, and $b \mid 10$.

Divisors.

Divisors of 3 : $\pm 1, \pm 3$,

Divisors of 10 : $\pm 1, \pm 2, \pm 5, \pm 10$.

Possible rational roots.

Forming all reduced fractions $\frac{a}{b}$ gives

$$\pm 1, \pm 3, \pm \frac{1}{2}, \pm \frac{3}{2}, \pm \frac{1}{5}, \pm \frac{3}{5}, \pm \frac{1}{10}, \pm \frac{3}{10}.$$

Evaluation of all candidates.

x	$f(x)$
1	30
-1	-12
3	900
-3	576
$\frac{1}{2}$	$\frac{225}{8}$
$-\frac{1}{2}$	$-\frac{27}{8}$
$\frac{3}{2}$	192
$-\frac{3}{2}$	0
$\frac{1}{5}$	$\frac{336}{125}$
$-\frac{1}{5}$	0
$\frac{3}{5}$	$\frac{144}{25}$
$-\frac{3}{5}$	$-\frac{96}{25}$
$\frac{1}{10}$	$\frac{147}{1000}$
$-\frac{1}{10}$	$-\frac{63}{1000}$
$\frac{3}{10}$	$\frac{459}{1000}$
$-\frac{3}{10}$	$-\frac{231}{1000}$

Since

$$f\left(-\frac{1}{5}\right) = 0 \quad \text{and} \quad f\left(-\frac{3}{2}\right) = 0,$$

and all other candidates give nonzero values, the rational roots are

$$\boxed{x = -\frac{1}{5} \quad \text{and} \quad x = -\frac{3}{2}}.$$

Problem 3

a) Consider the expression

$$\sqrt{14 + 6\sqrt{5}} - \sqrt{6 + 2\sqrt{5}}.$$

We rewrite each radical as a perfect square.

$$14 + 6\sqrt{5} = 9 + 5 + 2 \cdot 3\sqrt{5} = (3 + \sqrt{5})^2.$$

$$6 + 2\sqrt{5} = 1 + 5 + 2 \cdot 1\sqrt{5} = (1 + \sqrt{5})^2.$$

Therefore,

$$\sqrt{14 + 6\sqrt{5}} - \sqrt{6 + 2\sqrt{5}} = \sqrt{(3 + \sqrt{5})^2} - \sqrt{(1 + \sqrt{5})^2}.$$

Since both quantities are positive, this simplifies to

$$(3 + \sqrt{5}) - (1 + \sqrt{5}) = 2.$$

Thus, the value of the expression is

$$\boxed{2},$$

which is rational.

b) Suppose that

$$r = \frac{\log 2}{\log 3}$$

is rational. Then there exist integers $p, q > 0$ such that

$$r = \frac{p}{q}.$$

Thus,

$$\frac{\log 2}{\log 3} = \frac{p}{q}.$$

Multiplying both sides by $\log 3$, we obtain

$$\log 2 = \frac{p}{q} \log 3.$$

Multiplying both sides by q gives

$$q \log 2 = p \log 3.$$

Exponentiating both sides with base 10, we have

$$10^{q \log 2} = 10^{p \log 3}.$$

Using the identity $10^{\log a} = a$, this simplifies to

$$2^q = 3^p.$$

However, 2^q is a power of an even number, while 3^p is a power of an odd number. Since no power of an even integer can equal a power of an odd integer, this is a contradiction.

Therefore, our assumption was false, and

$$\boxed{\frac{\log 2}{\log 3} \text{ is irrational.}}$$

Problem 4

$$|x - 2| + |x - 4| < x + 1.$$

The critical points occur when the expressions inside the absolute values are zero:

$$x - 2 = 0 \quad \text{and} \quad x - 4 = 0,$$

so the critical points are $x = 2$ and $x = 4$.

We consider three cases.

Case 1: $x < 2$.

In this case, $|x - 2| = 2 - x$ and $|x - 4| = 4 - x$. Thus,

$$(2 - x) + (4 - x) < x + 1.$$

$$6 - 2x < x + 1,$$

$$-3x < -5,$$

$$x > \frac{5}{3}.$$

Hence,

$$x \in \left(\frac{5}{3}, 2\right).$$

Case 2: $2 \leq x < 4$.

Here, $|x - 2| = x - 2$ and $|x - 4| = 4 - x$. Thus,

$$(x - 2) + (4 - x) < x + 1,$$

$$2 < x + 1,$$

$$x > 1.$$

Since this holds for all x in the interval, we have

$$x \in [2, 4).$$

Case 3: $x \geq 4$.

In this case, $|x - 2| = x - 2$ and $|x - 4| = x - 4$. Thus,

$$(x - 2) + (x - 4) < x + 1,$$

$$2x - 6 < x + 1,$$

$$x < 7.$$

Hence,

$$x \in [4, 7).$$

Conclusion.

Combining all three cases, the solution set is

$$\boxed{\left(\frac{5}{3}, 7\right)}.$$

Problem 5

(a) To show that the set

$$S = \left\{ \frac{m}{n} : m, n \in \mathbb{N}, m < 3n \right\}$$

is bounded, we show that it is bounded above and bounded below.

Bounded below. Since $m, n \in \mathbb{N}$, we have $m \geq 1$ and $n \geq 1$, so

$$\frac{m}{n} > 0$$

for every element of S . Hence, 0 is a lower bound for S , and S is bounded below.

Bounded above. Since $m < 3n$, dividing both sides by $n > 0$ gives

$$\frac{m}{n} < 3.$$

Thus, every element of S is less than 3, so 3 is an upper bound for S , and S is bounded above.

Conclusion. Since S is bounded both above and below, S is bounded.

(b) **Finding $\inf(S)$ and $\sup(S)$.**

Infimum. From part (a), since $m, n \in \mathbb{N}$, we have $m \geq 1$ and $n \geq 1$, so

$$\frac{m}{n} > 0$$

for all $\frac{m}{n} \in S$. Hence, 0 is a lower bound of S .

To show that 0 is the greatest lower bound, let $\varepsilon > 0$ be arbitrary. Choose $m = 1$ and let $n \in \mathbb{N}$ be sufficiently large. Then

$$\frac{m}{n} = \frac{1}{n} \in S,$$

since $1 < 3n$ for all $n \in \mathbb{N}$.

By the Archimedean Axiom Corollary, there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < \varepsilon.$$

Thus, there exists $s \in S$ such that

$$0 < s < \varepsilon.$$

Since for every $\varepsilon > 0$ there exists $s \in S$ with

$$s < 0 + \varepsilon,$$

no number greater than 0 can be a lower bound of S . Therefore,

$$\inf(S) = 0.$$

Supremum. From part (a), since $m < 3n$, dividing both sides by $n > 0$ gives

$$\frac{m}{n} < 3$$

for all $\frac{m}{n} \in S$. Hence, 3 is an upper bound of S .

To show that 3 is the least upper bound, let $\varepsilon > 0$ be arbitrary. Choose $m = 3n - 1$ with $n \in \mathbb{N}$. Then $m < 3n$, and

$$s = \frac{m}{n} = \frac{3n - 1}{n} = 3 - \frac{1}{n} \in S.$$

By the Archimedean Axiom Corollary, there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < \varepsilon.$$

Thus,

$$s = 3 - \frac{1}{n} > 3 - \varepsilon.$$

Since for every $\varepsilon > 0$ there exists $s \in S$ such that

$$3 - \varepsilon < s < 3,$$

no number smaller than 3 can be an upper bound of S . Therefore,

$$\sup(S) = 3.$$